When Search Is Not Enough: Diagnosing Failure Before Constructing Verified Reasoning Capabilities

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Many reasoning systems respond to failure by generating more programs, refining candidates or adding test-time computation; these strategies are bounded by the representations and operations available to search. We study a complementary question: can a reasoner tell whether a failure reflects insufficient search, an excluded representation, a missing executable capability or insufficient evidence, and adapt only that component without compromising the evidence used to judge it?

We present CORA, a framework for failure-conditioned construction and verification of task-local executable extensions. A candidate is built from auditable failure traces, installed provisionally into the unchanged native reasoner, and checked by demonstration replay, adaptive leave-one-out rebuilt per fold, independent target evaluation and removal controls; abstention is a legitimate outcome.

A frozen synthetic experiment produced 27 of 30 complete end-to-end witnesses (12 of 30 for the earlier engine) within its declared grammar. On 60 real ARC-AGI-2 training-development tasks, four automatically constructed growth relations compiled and installed (none before), and one was accepted and used by the native engine. None survived adaptive leave-one-out (0 of 14 folds), so no new ARC solve is claimed. An instrumented failure funnel on the same 60 tasks, with held-out answers read only by an independent evaluator, then located where the native reasoner fails. Its leave-one-out verifier certified 6 answers, all correct; switching it off would add 2 correct and 26 wrong answers. In 27 tasks every program that fit the demonstrations was wrong: mostly memorized constant growth patterns, rebuilt differently by each leave-one-out fold. Neither the reasoner's own relational re-search nor three times the search budget repaired any of them, and the view-admission failure we had first traced by hand explains only 7 of 76 failed folds. The obvious view repairs were tested on synthetic controls first: forcing a view to be searched first gained no task and harmed five of 24, and a fold-local alternative gained one and harmed four of 41, so neither was adopted. Frozen neural guides chose views no better than the fixed order (Qwen2.5-1.5B 19 of 35, I-JEPA 3 of 35, pre-registered).

We then implemented the construction step (CORA-FHR): a declared grammar of typed growth operators, the reasoner's own memorized-growth record as evidence, reversible installation into the unchanged engine, and leave-one-out that rebuilds the operator in every fold. On 27 synthetic development tasks, one constructed single-step operator (a ring around an object's minority-colour cell; the task's planted second step adds nothing on its shapes), absent from the native and existing constructor vocabularies, was rebuilt in every fold and gave a correct certified prediction. There were no wrong certificates, and matched extra search gained nothing. A failure-blind constructor reached the same result, so the contribution of the failure record itself is not yet established; the prospective test is sealed.

We outline a domain-portable control protocol, whose general failure-type controller is not yet implemented, and a prospective biological-perturbation test (untested). Neither biological transfer nor real ARC invention has been demonstrated.

Keywords: ARC-AGI-2; program synthesis; failure diagnosis; capability construction; verification; leave-one-out.

Evidence tiers: synthetic (frozen, run once); real development (training demonstrations only, no test output read); forensic (subsets examined after the failure was seen). The certified training score is unchanged at 185/1,000, and the native solver's official Kaggle ARC-AGI-2 public score is 0.00 (October 2026). A separately labelled neural baseline (a reproduction of a public NVARC test-time-training pipeline, not CORA) scored 27.64 on the official public leaderboard (14 October), equal to NVARC's own 2025 public score; it is not a CORA result. A 120-task prospective pilot is frozen and has not been run. Every number is indexed with its commit and hash in `records/CORA_SUMMIT_POSTER_EVIDENCE_20261012.md` and `records/CORA_FAILURE_FUNNEL_20261014.md`.